Journal of Molecular Liquids 324 (2021) 114779

Contents lists available at ScienceDirect

Journal of Molecular Liquids

journal homepage: www.elsevier.com/locate/molliq

Thermal behavior, solvatochromic parameters, and metal halide solvation of the novel water-based deep eutectic solvents

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ARTICLE INFO

Article history:
Received 23 June 2020
Received in revised form 22 September 2020
Accepted 11 November 2020
Available online 17 November 2020

Keywords:
Water
Deep eutectic solvent
Solvatochromism
Thermal behavior
Metal halides
Solvation

ABSTRACT

High viscosity of a deep eutectic solvent (DES) is one of the biggest challenges to their commercial use. In this study, two novel water-based low-viscosity DESs were formulated and could be easily applicable in many areas of research and industry, such as pharmaceuticals, waste management, and biorefinery. To formulate the DESs, water was used as a sole hydrogen bond donor, while the hydrogen bond acceptor was choline chloride. Melting points of 1:3 and 1:4 choline chloride: H_2O were found to be significantly low, −79.21 and − 79.25 °C, respectively. TGA showed the stability of the DES over a wide temperature range. For example, in the TGA thermogram of 1:3 ChCl: H_2O, was found to be crystalline up to 148.64 °C and the onset point was 193.75 °C. The solvent selectivity triangle from Kamlet-Taft parameters proved that the DESs possess similar solvatochromic properties to ionic liquids. Therefore, solvatochromism helped to understand the solvation of the DESs. A simple analytical method was developed employing ion chromatography and atomic absorption spectroscopy to investigate the solubility of sodium halides, alkali chlorides, and cobalt chloride in the studied DESs. Solubility trends of the metal halides in both DESs were NaCl > NaBr > NaI > NaF for sodium halides and LiCl > NaCl > KCl for alkali metal chlorides. Among all the studied metal halides, solubilities of the $CoCl_2$ and LiCl were the highest. These findings indicate that the formulated DESs could be a good metal extraction media for the recycling process of spent Li-ion battery.

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1. Introduction

Use of greener chemicals, development of sustainable technologies, and industrial compliance with regulations are the tools to build our world safer. Alternative solvents could play a major role to make the dream in reality [1]. Organic solvents have been used for centuries, which are typically toxic to the environment. In the last few decades, some greener alternative solvents have been emerged, and among them ionic liquids (ILs) show excellent qualities. Although ILs show great promise, their expense and toxicity make it difficult to use them on a large scale [1–5]. Deep eutectic solvents (DESs) has been emerged in the past two decades to overcome these drawbacks [6–10]. Deep eutectic solvents show promise in various fields, such as chemistry, material science, physics, and pharmaceutical science. It has the advantages of being nonflammable with negligible vapor pressure compared with molecular solvents and share many beneficial qualities of ILs [11]. DESs are also cheaper to formulate, nontoxic, highly pure, easily recyclable, and biodegradable [12]. Although DESs are environmentally benign and considered as one of the most versatile alternative media, detailed knowledge of their miscibility, polarity, and solvation behavior are yet to be explored. High viscosity and disruption of the eutectic environment by water ingestion cause major problems to their use on a large scale [13]. Recent studies reported that water has some unique effects on the physicochemical behavior and chemical structures of DES. If water is added to a DES system at a high mole fraction (e.g., >25%), the HBA and HBD hydrate and show higher diffusivity, and the water molecules dominate the hydrogen bond formation with the constituents of the DES which eventually dampens the DES system [14]. Another study strengthened the claim that the water molecules make stronger interactions to the components of a DES than the interactions among themselves (e.g., water-water) or among the DES-DES clusters [15]. Viscosity of a DES typically decreases dramatically with the addition of a small amount of water. For instance, Savi et al. quantified that when approximately 10% of water is added to the 5:1 lactic acid:glucose DES, viscosity of the DES decreases to 85.9% [16]. Therefore, it is necessary to understand the role of water in DES.

We formulated a novel type of DES where water is used as a sole hydrogen bond donor. These DESs can be called as water-based DESs (WDESs). To date, only Marcus [17] claimed that some aqueous salt hydrates should be considered as DES. Asare [18] demonstrated that water can be a sole hydrogen bond donor (HBD) to form DES with ChCl. In this study, we have focused only on the combinations of 1:3 (WDES1) and 1:4 ChCl:H_2O (WDES2).

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https://doi.org/10.1016/j.molliq.2020.114779
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Thermal behavior of a DES is crucial to know because it helps to estimate the range of its applicability [19]. To measure the eutectic point and thermal stability of the solvents, differential scanning calorimetry (DSC) and thermal gravimetric analysis (TGA) techniques have been widely used [20]. Solvatochromism can play an important role to select a proper alternative solvent for a specific application [21]. Similar to an organic solvent, a DES system is hard to describe only by dielectric constant or dipole moment because it possesses many interactions, such as the double ionic effect, hydrogen bonding, halogen bonding, π-π interactions, and van der Waals forces [22–24]. Since most of the available methods allow examination of only one major type of interaction, a range of different probes are required to get a clear insight of the strength of solute-solvent interactions in a DES system. Solvatochromic measurements can determine some important properties, such as transition energy (E_T^N), hydrogen bond donor acidity (α), hydrogen bond acceptor basicity (β), and polarizability (π*) of such binary solvents at the same time through measuring the solvatochromic responses of several ultraviolet-visible (UV–Vis) absorbance probes [22,25–27].

Solvatochromism measures the interactions involved with solvation of dyes with the DESs, but solubility of ionic compounds in DESs has been rarely investigated. Few works have been published based on metal oxide and sodium salt solubility in some DESs using two methods, e.g. inductively coupled plasma-atomic emission spectroscopy (ICP-AES) [28,29] and shake-flask methods [29]. In this study, a simple solubility determination technique has been developed using ion chromatography (IC) and flame atomic absorption spectroscopy (FAAS) to determine the solubility of sodium halides, alkali chlorides, and $CoCl_2$ in WDESs. Besides solvatochromism, solubility results can help to understand the extractability of the WDESs. Several studies on metal extraction, such as Li and Co extraction from Li-ion battery waste [30], extraction of Co and Fe from NdFeB magnets [31], dissolution and recovery of electrum, native Te, tellurobismuthite, galena, and chalcopyrite [32], and extraction of metal salts and oxides from incinerated sewage sludge and ash [33] using DESs have been reported. Therefore, it has already been established that certain types of DESs could be the excellent extraction media for the specific metal recycling process. Therefore, the aim of the solubility study of the metal halides is to investigate the primary data of the Li and Co extractability of the WDESs.

The overall objectives of this study are (i) to understand the difficulties related to high viscosity of DES and to answer the questions, such as why water disrupt or disturb a eutectic environment or can water be a sole hydrogen bond donor (HBD) to form a DES?, (ii) if water can form a DES with an HBA, then how do the nonbonding interactions influence to decrease the melting points and to increase their thermal stabilities?, (iii) how do the solvatochromic parameters and solvation behavior of a WDES correlates to the selected ionic liquids?, (iv) do the solubilities of metal halides in water decrease in WDESs?, (v) to get preliminary data on the extractability of Co^{2+} and Li^+ ions from spent Li-ion battery.

2. Experimental details

2.1. Materials

Nile Red, 4-nitroaniline, and dichloromethane (DCM) were purchased from Thermo-Fischer Scientific (Dubuque, IA). Choline chloride (ChCl), cobalt chloride ($CoCl_2$), sodium fluoride (NaF), sodium chloride (NaCl), sodium bromide (NaBr), sodium iodide (NaI), lithium chloride (LiCl), and methanol were purchased from Acros Organics. Potassium chloride (KCl) and N,N-diethyl-4-nitroaniline were purchased from Fisher Chemicals and Frinton Laboratories, respectively. All chemicals have purity ≥98.0%. Ultrapure water was used for the solubility measurements with metal halides.

2.2. Formulation of the WDESs

WDESs were formulated by mixing anhydrous choline chloride and water at 1:3 and 1:4 M ratios in a covered beaker with continuous stirring (600 rpm) for 30 min under atmospheric temperature and pressure until homogenous liquids were obtained. Solvents were monitored for one hour to confirm if there were any crystals left at the bottom of the beakers. After the preparation, solvents were stored in a desiccator.

2.3. Differential scanning calorimetry and thermogravimetric analysis method

Melting temperatures (T_m) of the WDESs were measured using a TA-Q2000 differential scanning calorimeter at a heating rate of 10 °C per min. About 10 mg of sample was weighed into an aluminum hermetic pan. An empty reference aluminum pan was kept under the same conditions throughout the experiment. Both pans were covered with premium aluminum lids. DSC measurements for the WDESs were conducted over the temperature range of −80 °C to 100 °C. The advanced Tzero™ technology of the calorimeter controls temperature throughout the experiment because its thermocouple sensor ensures accurate heat flow into the samples by considering any asymmetries. Nitrogen gas was purged continuously at the rate of 20 mL per minute to avoid moisture contamination. Accuracy and reproducibility of the DSC measurements were ensured by the regular calibration of the instrument with an indium metal standard.

The heat tolerance of the DESs were analyzed using thermogravimetric analysis (TGA) method. TGA was conducted using Seiko 220 TG/DTA (Tokyo, Japan) instrument. A sample size of about 10 mg was placed in an aluminum hermetic pan. The samples (DESs) were heated from 30 to 300 °C at a temperature ramp of 10 °C per min under nitrogen atmosphere.

2.4. Solvatochromism

Solvatochromic measurements were carried out using three widely used dyes, Nile red, 4-nitroaniline, and N, N-diethyl-4-nitroaniline (Fig. 1). Individual solutions of 10^{-3}, 10^{-4}, 10^{-5}, and 5×10^{-4} M (M) of the three dyes were prepared in dichloromethane (DCM). Among the four concentrations, 5×10^{-4} M solution worked best for the WDESs. An appropriate amount of the dye stock solution was micropipetted and transferred to the clean and dry glass vials to mix with WDES1, WDES2, and methanol (control solvent). The glass vials were then capped, vortexed, and mixed well to get the homogeneous solutions. A 200 μL aliquot of each sample was loaded into a 96-well plate.

A UV–Vis spectrophotometer (Synergy H1 microplate reader, Biotek) was used to determine the wavelength of maximum absorbance (λ_{max}) of the dyes at ambient temperature. Absorbance readings (triplicate) were collected by spectral scanning of WDESs-dye solutions at 1 nm interval. All the solvatochromic properties of WDES1 and WDES2 were calculated using the following equations, where, υ α, β, π*, and E_T(NR) refer to the wavenumber, hydrogen bond donor acidity, hydrogen bond acceptor basicity, polarizability, and transition energy, respectively. E_T^N is the dimensionless transition energy.

$$E_T(NR)_{kcal.mol^{-1}} = \frac{28591.44}{\lambda_{max}(nm)} \quad (1)$$

$$\pi* = 0.314\left(27.52 - \upsilon_{N,N-diethyl-4-nitroaniline}\right) \quad (2)$$

$$\alpha = 0.0649\, E_T\,(NR) - 2.03 - 0.72\,\pi* \quad (3)$$

$$\beta = \frac{1.035\,\upsilon_{N,N-diethyl-4-nitroaniline} + 2.64 - \upsilon_{4-nitroaniline}}{2.80} \quad (4)$$

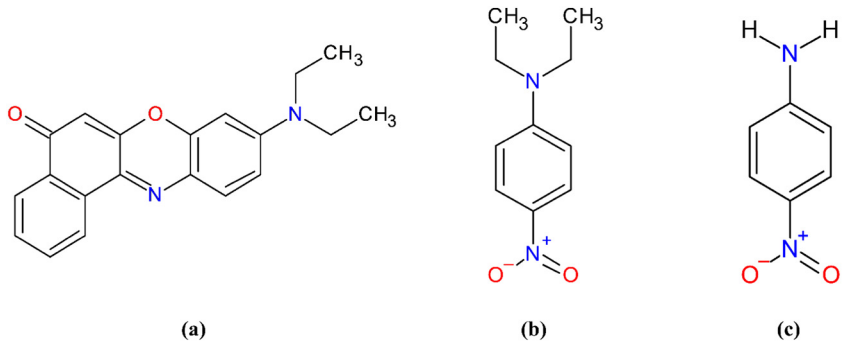


Fig. 1. Two dimensional (2D) structures of the solvatochromic dye molecules used in this study. (a) Nile Red, (b) N, N-diethyl-4-nitroaniline, and (c) 4-nitroaniline.

$$E_T^N = \frac{E_T(solvent) - E_T(TMS)}{E_T(water) - E_T(TMS)} \quad (5)$$

2.5. Ion chromatography and atomic absorption spectroscopic method

To prepare saturated solutions of alkyl chlorides, sodium halides, and cobalt chloride ($CoCl_2$) in the WDESs, 100 mg of each metal halide was added to five mL of each WDES in a small covered beaker with continuous stirring at 600 rpm. If necessary, a small amount of WDESs was continuously added until a clear solution was obtained. This solvation process was monitored about 48 h to get an approximate solubility of the metal halides in the WDESs. Some of the metal halides co-crystallized during the solvation period if the stirring and salt addition continued over 24 h. Hence, the preparation of the saturated solutions were performed for 24 h in all cases. To get the true solubility of the metal halides, a simple method was developed using ion chromatography (IC) and flame atomic absorption spectroscopy (FAAS). From the initial study, we determined the approximate amount of metal halides required for the preparation of the saturated solutions. Therefore, to prepare samples for the analytical methods, we started with enough amount of metal halides in 10 mL of WDESs and stirred continuously for 24 h. Saturated solutions were filtered using 0.2 μm filter and transferred to the glass vials. Exactly 0.1 mL of the saturated solution was micropipetted and dissolved in 1000 mL of ultrapure water and then two mL (except LiCl, one mL was taken for the second dilution) of that diluted solution was taken and dissolved in another 100 mL ultrapure water in a volumetric flask to do second dilution. Sample preparation for the AAS needed additional acid digestion step.

Prepared samples were analyzed using a Thermo Scientific Dionex ICS 1500 ion chromatography system. An IonPac CS12 (2 × 250 mm) column was used, and the eluent was 12 mM H_2SO_4. Injection volume and the flow rate were 1.0 mL and 0.50 mL/min, respectively. Detection mode of the instrument is suppressed conductivity using a CERS 500 2-mm suppressor. PinAAcle™ 500 (Perkin Elmer) a corrosion-resistant flame atomic absorption spectrometer (FAAS) was used to analyze $CoCl_2$.

3. Results and discussion

3.1. Thermal behavior of the WDESs

Differential scanning calorimetry (DSC) measures heat flow associated with the thermal transitions. Melting points of the WDESs are obtained through the endothermic process and these were found at −79.21 ± 0.28 °C for WDES1 and − 79.25 ± 0.32 °C for WDES2 (Fig. 2). These results are cross-validated by keeping the WDESs in different freezers within the temperature range between −65.0 to −85.0 °C. The depression of freezing point refers to the nature and extent of the intermolecular interactions among the molecules of HBA and HBD. A variety of interactions, such as electrostatic interaction, hydrogen bonds, halogen bonds, and van der Waals forces, among the molecules cause the extreme depression of the melting points of the WDESs.

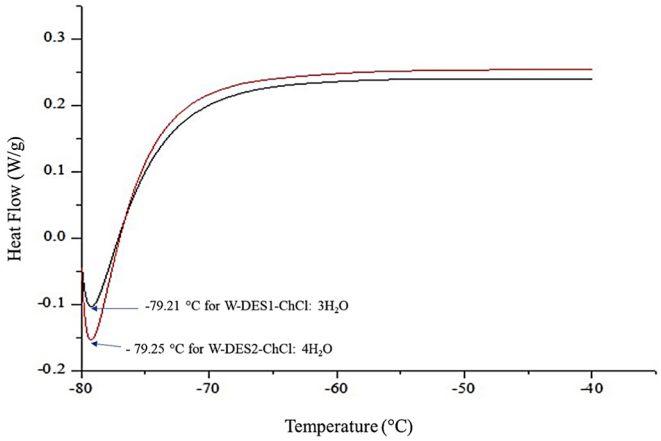


Fig. 2. DSC curves of WDES1 and WDES2.

Stability of a solvent at high temperature refers to the strength of the interactive forces among the solvent molecules. Same intermolecular interactions stated in the DSC study make a solvent stable at high temperature. Figs. 3 displays the TGA thermogram of WDES1 and WDES2. The major physical transitions in the water molecules are observed at 193.75 and 197.73 °C, respectively. Crystalline water is found up to 148.64 °C for WDES1 and 116.13 °C for WDES2. High pressure in the capped aluminum pan and surrounding enclosure facilitate determination of the exact transition points at high temperature. These results justify high thermal stability of the water molecules when they form WDESs with choline chloride.

3.2. Solvatochromic parameter analysis

Solvatochromic parameters, including $E_T(NR)$ or E_T^N, α, β, and π*, of the WDESs indicate that their properties are similar to ionic liquids [34]. This is due to the comparable intermolecular interactions between the WDESs and the dye molecules. Hydrogen bonds, electrostatic interactions or double ionic effect, halogen bonds, and van der Waal forces are the major interactions in the solutions. In this experiment, methanol is used as a control solvent. Solvatochromic parameters measured for methanol are close to the literature values. Reasons for deviation between the measured and literature values could be due to dissolved gasses in the sample solutions and the moisture present in the dyes used for the solvatochromic measurements [35–38]. The transition energy and polarizability of the methanol are found 51.98 kcal.mol^{-1} and

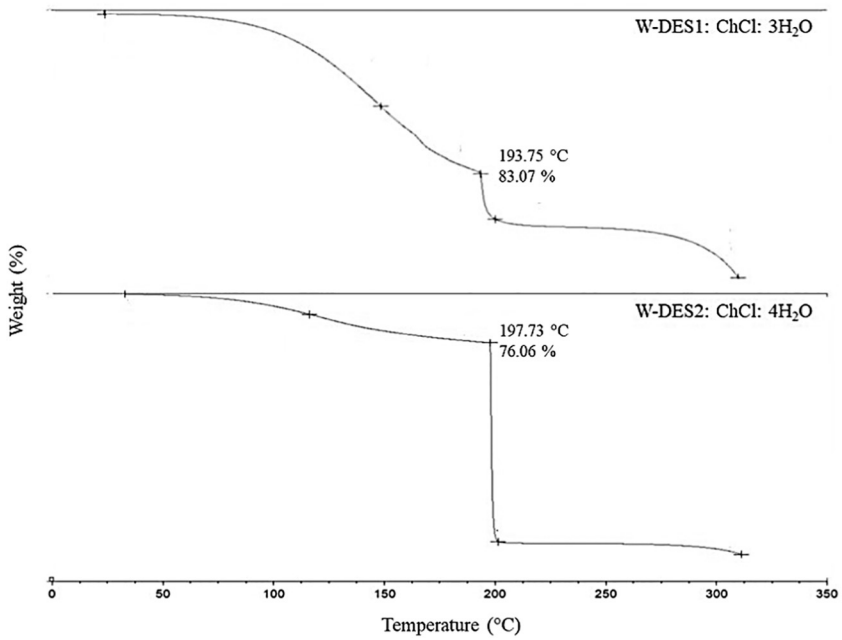


Fig. 3. Thermogravimetric analysis of WDES1 and WDES2.

0.79, while the literature values are 52.02 kcal. Mol^{-1} [39] and 0.77 [40], respectively. The dimensionless E_T^N values of the WDES1 and WDES2 are 0.61 and 0.62, respectively at ambient temperature. Low transition energy values of the WDESs represent high polarity of the solvents. Polarity values are 1.1615 and 1.0014 for WDES1 and WDES2, respectively.

To compare the measured Kamlet-Taft parameters (α, β, and π*) of the WDESs, same parameters of the selected organic solvents and some common ionic liquids care compared in Table S1. To visualize the comparison, a ternary plot was constructed using Origin Pro 2020 software package [41], presented in Fig. 4. This ternary plot is called a Snyder solvent selectivity triangle (SST), which characterizes the solvents depending on their Kamlet-Taft parameters and makes clusters or groups of them by plotting α, β, and π* of the solvents in a triangle [42,43]. The triangle shows that the WDESs are closely related to ILs, but away from the organic solvents. That means formulated DESs possess similar solvatochromic properties to IL, which indicates that they could be excellent extraction media for a variety of compounds. To get deeper insight of the analysis, another ternary plot or SST (Fig. S1) is plotted to compare the solvatochromic parameters of WDESs with only the ILs. Groups of the ILs are determined based on their cations to investigate if there is any correlation among the structural features of the WDESs and ILs. MBP-based ILs are located most close to the location of the WDESs, it could be because of both Ch^+ and MBP^+ have only one N atom (specifically one N^+) in their structures. Except them, all the other cations of the listed ILs have one N atom and one N^+ cation in their structures. HOEmim-based ILs also closely orients around the WDESs maybe because of both the Ch^+ and $HOEmim^+$ have the N^+-CH_2-CH_2-OH group. Structures of the other cations are similar to the HOEmim except the -OH functional group, and most of the selected ILs (Table S1) are clustered around the WDESs. Factors, such as types of anions, structural size, number of HBA and HBD atoms, number of rotatable, single, double, and hydrogen bonds could be the reasons behind the few anomalous results.

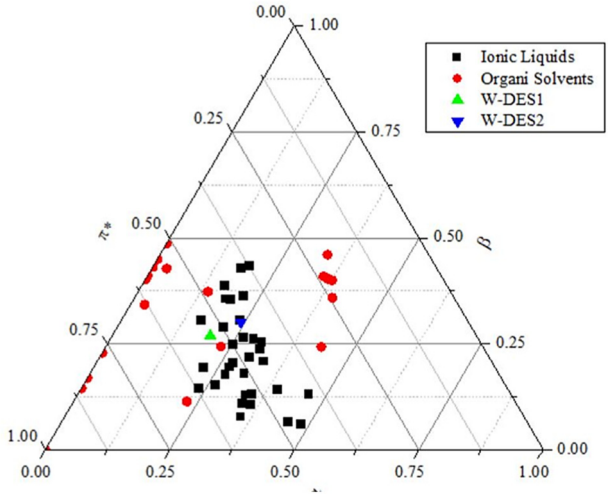


Fig. 4. Ternary plot of Kamlet-Taft parameters (α, β, and π*), where the parameters of the WDESs were compared to some selected organic solvents and ionic liquids.

3.3. Metal halide solubility in WDESs

Since DES is a complex and viscous solvent, it is difficult to measure the solubility of different compounds in it. In this study, a simple

Table 1
Solubility of the listed metal halides in water and the WDESs at ambient temperature.

	Solubility in Water and WDESs (g/100 mL)*		
Metal Halides	Water	WDES1	WDES2
Sodium Halides Solubility (g/100 mL ± SDE)			
NaF	4.3[44]	1.616 ± 0.006	1.260 ± 0.015
NaCl	36.0[45]	6.993 ± 0.068	5.861 ± 0.076
NaBr	94.6[46]	4.971 ± 0.055	3.407 ± 0.041
NaI	184.0[47]	3.634 ± 0.021	2.829 ± 0.032
Alkali Chlorides Solubility (g/100 mL ± SDE)			
LiCl	84.5[48]	8.388 ± 0.122	15.203 ± 0.238
NaCl	36.0[45]	6.993 ± 0.051	5.861 ± 0.031
KCl	35.5[49]	0.244 ± 0.014	0.257 ± 0.014
Cobalt Chlorides Solubility (g/100 mL ± SDE)			
$CoCl_2$	53.0[50]	20.981 ± 0.174	23.895 ± 0.301

* SDE means standard error.

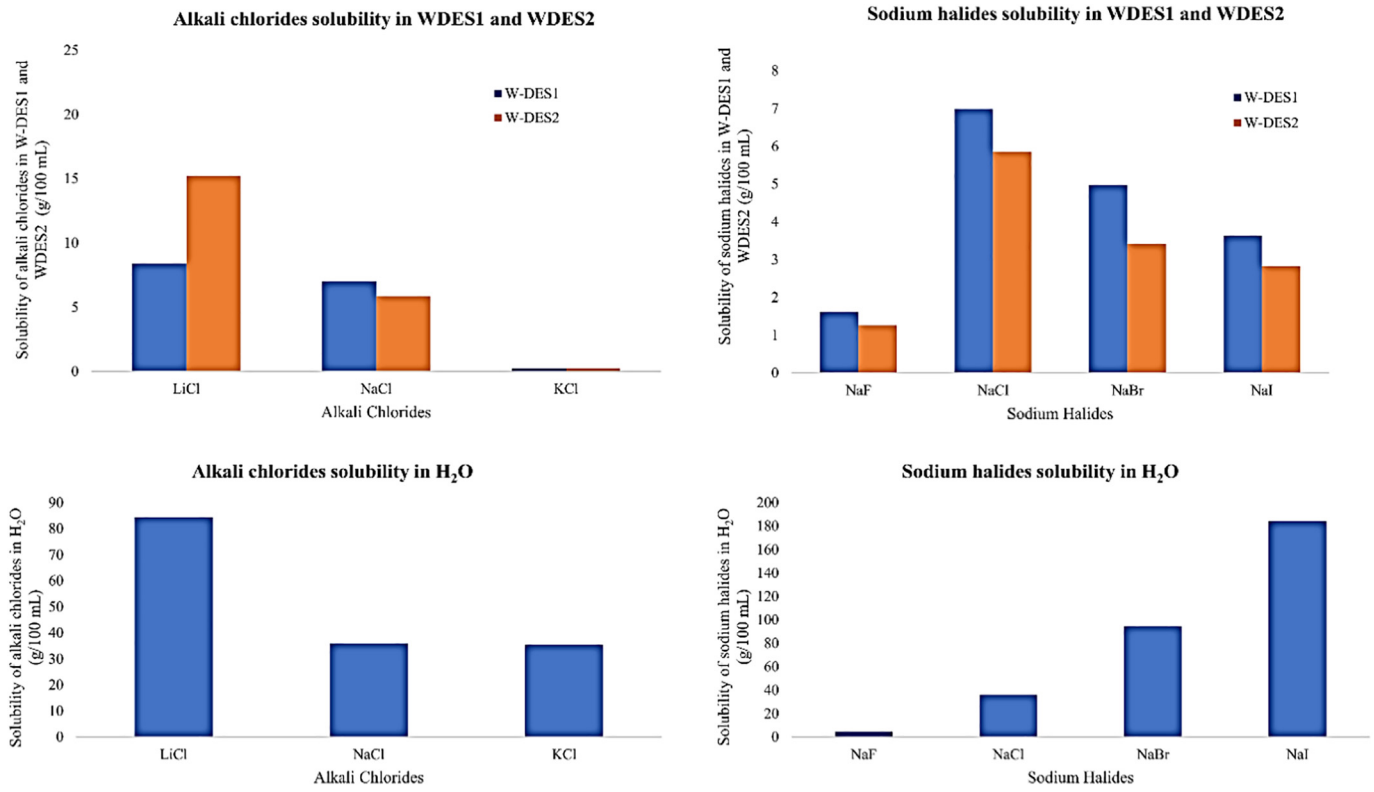


Fig. 5. Solubility trends of alkali chlorides and sodium halides in water and the WDESs.

method is developed to determine the solubility of the metal halides in the WDESs under ambient temperature and pressure. Solubility of the sodium halides and alkali chlorides in the WDESs are determined using ion chromatography and $CoCl_2$ is analyzed by FAAS. Table 1 and Fig. 5 show the solubility of the metal halides studied in the water-based deep eutectic solvents.

It is observed that solubilities of the studied metal halides decreased considerably in the WDESs compared to their solubilities in water. The major reason behind the phenomena could be ion pair repulsion between choline chloride and the metal halides. Solubility trends of sodium halides and alkali chlorides in water are NaI > NaBr > NaCl > NaF and LiCl > > NaCl > KCl, respectively. But in the WDESs, solubility trends are NaCl > NaBr > NaI > NaF for sodium halides and LiCl > NaCl > KCl for alkali metal chlorides.

It is also found that LiCl and $CoCl_2$ are the two most soluble metal halides in the WDESs. It happens because the atomic size of the both cations are so small that they can easily diffuse through the gaps the solvent systems. DES (e.g., ChCl: ethylene glycol) is an efficient extraction solvent for the two major metals (Li and Co) [30] of spent Li-ion batteries. DESs are proved media to recycle other crucial metals (e.g., Fe and Te) and metal compounds [31–33]. Since our developed WDESs are inexpensive, nonflammable, more stable across a range of temperatures, and efficient to solubilize Li and Co salts, they could be good solvents to extract Li and Co metals from the waste of Li-ion batteries.

4. Conclusions

In summary, water-based deep eutectic solvents are sustainable and environment friendly. They are easier to formulate, cheaper, and safer for a wide range of applications. Thermochemical studies proved that the WDESs are applicable from extremely low temperature to high temperature. Although solubilities of the metal halides were decreased compared to their solubilities in water, their high polarizability values indicate that they could increase the solubilities of a wide range of compounds (e.g., active pharmaceutical ingredients). Besides, the WDESs could selectively extract some important metals (e.g., Li and Co) or metallic compounds from the waste materials, such as spent Li-ion batteries. The studies of their physicochemical properties will certainly help to understand the role of water molecules in the formation and disruption of a DES.

Notes

The authors declare no competing financial interest.

CRediT authorship contribution statement

Md Sajjadur Rahman: Conceptualization, Writing - review & editing, Validation. **Douglas E. Raynie:** Conceptualization, Validation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Acknowledgements

The authors are grateful to the Department of Chemistry and Biochemistry, South Dakota State University (SDSU) for their overall facilities. Besides authors are also grateful to Dr. Jihong Cold-Dai and Carleigh Larrick, Department of Chemistry and Biochemistry, SDSU to allow and help us running samples in their ion-chromatography instrument. The authors also would like to thank Dr. Regina Wixon

and Aubrey Jones from SDAgLab for allowing and helping us to run samples in their Flame Atomic Absorption Spectroscopy instrument. The authors are grateful to the Department of Agricultural and Biosystem Engineering, SDSU for giving the permission to use their DSC and TGA instruments.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2020.114779.

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